



# DUNMAN SECONDARY SCHOOL

CANDIDATE  
NAME

CLASS

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## PRELIMINARY EXAMINATION 2023 SECONDARY 4 NORMAL ACADEMIC

**MATHEMATICS SYLLABUS A**

Paper 2

**4045/02**

31 July 2023

**2 hours**

# Marking Scheme

| Question | Answer                               |
|----------|--------------------------------------|
| 1a       | $2^2 \times 3 \times 5^2 \times 7$   |
| 1bi      | $2^3 \times 3^2 \times 5^2 \times 7$ |
| 1bii     | $3^2 \times 7$                       |

| Question | Answer                                                                              |
|----------|-------------------------------------------------------------------------------------|
| 2a       | $\angle ABC = \angle DEF$<br>$\angle ACB = \angle DFE$<br>$\angle BAC = \angle EDF$ |
| 2bi      | $\frac{BC}{9.45} = \frac{8.1}{14.175}$ $BC = 5.4 \text{ cm}$                        |
| 2bii     | $\frac{8.1}{14.175}$ $= \frac{4}{7}$                                                |

| Question | Answer                                                                                                                               |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| 3        | Percentage of adults who took between 100 and 120 minutes                                                                            |
|          | $= \frac{18}{5+6+9+22+15+18} \times 100\%$                                                                                           |
|          | $= 24\%$                                                                                                                             |
|          | <b>Not all 18 adults</b> who lie in the interval between 100 and 120 minutes <b>may have taken 120 minutes to complete the race.</b> |

| Question           | Answer                                                        |
|--------------------|---------------------------------------------------------------|
| 4ai                | $21 = k(6)^2$                                                 |
|                    | $k = \frac{7}{12}$                                            |
|                    | $y = \frac{7}{12}x^2$                                         |
| 4aai               | $y = \frac{7}{12}(9)^2$                                       |
|                    | $= 47.25$                                                     |
| 4b                 | $2x + y = 1$                                                  |
|                    | $y = 1 - 2x$ ----- (1)                                        |
|                    | $3x - 4y = 29$ ----- (2)                                      |
|                    | Sub. (1) into (2),                                            |
|                    | $3x - 4(1 - 2x) = 29$                                         |
|                    | $x = 3$                                                       |
|                    | $y = 1 - 2(3)$                                                |
| 4c                 | $y = -5$                                                      |
|                    | $5(3x + 1) - 4(9 - x) = 140$                                  |
|                    | $15x + 5 - 36 + 4x = 140$                                     |
|                    | $19x = 171$                                                   |
| Alternative method | $x = 9$                                                       |
|                    | $\frac{3}{4}x + \frac{1}{4} - \frac{9}{5} + \frac{1}{5}x = 7$ |
|                    | $\frac{19}{20}x = \frac{171}{20}$                             |
|                    | $x = 9$                                                       |

| Question | Answer                    |
|----------|---------------------------|
| 5a       | $6h^2 + 10h - h - 4h^2$   |
|          | $= 2h^2 + 9h$             |
| 5b       | $= 8(9m^2 - 4n^2)$        |
|          | $= 8(3m - 2n)(3m + 2n)$   |
| 5c       | $ab^3 - b^3 = 2a$         |
|          | $ab^3 - 2a = b^3$         |
|          | $a(b^3 - 2) = b^3$        |
|          | $a = \frac{b^3}{b^3 - 2}$ |

| Question | Answer                                                               |
|----------|----------------------------------------------------------------------|
| 6a       | $OQ^2 = 9^2 + 12^2$                                                  |
|          | $OQ = \sqrt{225}$                                                    |
|          | $OQ = 15 \text{ cm}$                                                 |
| 6b       | $\tan \angle XOQ = \frac{9}{12}$                                     |
|          | $\angle XOQ = 36.8698^\circ$                                         |
|          | $\angle POQ = 36.8698^\circ \times 2$                                |
|          | $= 73.7397^\circ$                                                    |
|          | $= 73.7^\circ (1 \text{ d.p.})$                                      |
| 6c       | Length of arc $PQ = \frac{73.7397^\circ}{360^\circ} \times 2\pi(15)$ |
|          | $= 19.305 \text{ cm}$                                                |
|          | Length of arc $PA = \frac{1}{4} (2\pi(18))$                          |
|          | $= 9\pi \text{ cm}$                                                  |
|          | Perimeter $= 19.305 + 9\pi + 18$                                     |
|          | $= 65.579 \text{ cm}$                                                |
|          | $= 65.6 \text{ cm} (3 \text{ s.f.})$                                 |

| Question           | Answer                                                   |
|--------------------|----------------------------------------------------------|
| 7a                 | Lina is cycling at a <b>constant speed of 7 m/s.</b>     |
| 7b                 | $\frac{4}{20}$                                           |
|                    | = 0.2 m/s                                                |
| 7c                 | $\frac{1}{2} \times (t - 90) \times 7 = 42$              |
|                    | $t = 102$                                                |
| Alternative method | $\frac{1}{2} \times \text{base} \times 7 = 42$           |
|                    | Base = 12                                                |
|                    | $t = 90 + 12$                                            |
|                    | $t = 102$                                                |
| Question           | Answer                                                   |
| 8a                 | $\frac{2}{3} \pi (8)^3 + \frac{1}{3} \pi (8)^2 (23 - 8)$ |
|                    | $= \frac{1984}{3} \pi$                                   |
|                    | = 2077.63                                                |
|                    | = 2080 cm <sup>3</sup> (3 s.f.)                          |
| 8b                 | $XY = \sqrt{8^2 + 15^2}$                                 |
|                    | $XY = 17 \text{ cm}$                                     |
| 8c                 | $2\pi(8)^2 + \pi(8)(17)$                                 |
|                    | = $264\pi$                                               |
|                    | = 829.38                                                 |
|                    | = 829 cm <sup>2</sup> (3 s.f.)                           |
| Question           | Answer                                                   |
| 9a                 | 8.5                                                      |

9b



9c

6.8

9d

balanced tangent drawn at  $x = 2$ Gradient =  $-2.75$ **Question****Answer**

10a

$$I = 11200 \left( 1 + \frac{2.65}{100} \right)^3 - 11200$$

= \$914.20 (nearest cent)

|       |                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------|
| 10bi  | $914.20 = \frac{P \times 3.48 \times 1}{100}$                                                                        |
|       | $P = \$26270$ (nearest dollar)                                                                                       |
| 10bii | $\$3550(12) = \$42600$<br>$< \$45000$<br>Should not choose Brown Bank as does not meet annual income criteria.       |
|       | Since maximum loan amount for Alpha and Central Bank is at least $\$3550(6) = \$21300$ , he can loan entire \$21000. |
|       | Interest to be paid to Alpha Bank<br>$= \frac{21000 \times 3.87 \times 0.5}{100}$<br>$= \$406.35$<br>$< \$914.20$    |
|       | Interest to be paid to Central Bank<br>$= \frac{21000 \times 3.48 \times 1}{100}$<br>$= \$730.80$<br>$< \$914.20$    |
|       | Since interest to be paid to Alpha Bank is the lowest,<br>Choose Alpha Bank with loan period of 0.5 months.          |

| Question | Answer                                                                             |
|----------|------------------------------------------------------------------------------------|
| 11a      | $\angle CAD = 30^\circ$                                                            |
|          | <b>Angles in the same segment are equal</b>                                        |
|          | Reflex $\angle BOD = 360^\circ - 102^\circ$<br>$= 258^\circ$                       |
|          | $\angle BCD = 258^\circ \div 2$<br>$= 129^\circ$                                   |
|          | <b>Angle at centre is twice the angle at circumference</b>                         |
| 11bi     | $\angle PQR = \cos^{-1} \left( \frac{245^2 - 182^2 - 130^2}{-2(182)(130)} \right)$ |
|          | $= 102.201^\circ$                                                                  |
|          | Bearing of $P$ from $Q = 115^\circ + 102.201^\circ$                                |
|          | $= 217.2^\circ$ (1 d.p.)                                                           |
| 11bii    | Let angle of elevation be $x$ .                                                    |
|          | $\tan x = \frac{45}{182}$                                                          |
|          | $x = 13.888^\circ$                                                                 |
|          | $x = 13.9^\circ$ (1 d.p.)                                                          |

*–1m from entire question if no degree symbol written at all*  
*–1m from entire question if answers of angles not in 1 d.p. or better*

| Question           | Answer                                                                                                                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| 12ai               | <b>Mean</b> = $\frac{11(550) + 15(650) + 26(750) + 45(850) + 33(950)}{130}$                                            |
|                    | = 806.92                                                                                                               |
|                    | = <b>807 g</b> (3 s.f.) ( <i>accept</i> $806\frac{12}{13}$ )                                                           |
|                    | <b>Standard deviation</b> =                                                                                            |
|                    | $\sqrt{\frac{11(550)^2 + 15(650)^2 + 26(750)^2 + 45(850)^2 + 33(950)^2}{130} - \left(806\frac{12}{13}\right)^2}$       |
|                    | = 122.12<br>= <b>122 g</b> (3.s.f.)                                                                                    |
| 12aii              | <b>Farm A</b> as its <b>standard deviation at 122 g</b> is <b>lower than</b> that of Farm B at <b>134 g</b> .          |
| 12bi               | $\frac{78}{130} \times \frac{77}{129}$                                                                                 |
|                    | = $\frac{77}{215}$                                                                                                     |
| 12bii              | $\frac{26}{130} \times \frac{25}{129} + \frac{26}{130} \times \frac{104}{129} + \frac{104}{130} \times \frac{26}{129}$ |
|                    | = $\frac{233}{645}$                                                                                                    |
| Alternative method |                                                                                                                        |
|                    | $1 - \left(\frac{104}{130} \times \frac{103}{129}\right)$                                                              |
|                    | = $\frac{233}{645}$                                                                                                    |